

Aloysius Allen Darmawan

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EDUCATION

Monash University

Feb. 2022 – Jul. 2026

Bachelor of Science (Pure Mathematics, Physics)/Bachelor of Computer Science

Melbourne, VIC

- Weighted Average Mark (WAM): Distinction average
- Completed study abroad semesters at National University of Singapore, Singapore and Lund University, Sweden

EXPERIENCE

Software Developer Intern

Jun. 2025 - Jan. 2026

Geotesta

Melbourne, VIC

- Developed a PyTorch image classifier for tunnel face rock structure recognition, integrating Google Vision and REST APIs via FastAPI, achieving 90% validation accuracy and reducing manual review time by 30%
- Deployed a full-stack website application for tailings dam safety analysis with an integrated AI chatbot, built using Flask and Bootstrap, streamlining workflows for 20+ engineers across multiple firms

Backend Engineer Intern

Nov. 2024 – Feb. 2025

ANZ Plus

Melbourne, VIC

- Streamlined CI/CD pipelines by building GitHub Actions workflows for release tagging and deployment validation across 5+ environments, reducing deployment errors by 40%
- Developed and deployed Go cloud-native microservices for real-time credit decisioning, automating SQL generation to query BigQuery datasets with 100M+ records resulting in scalable, data-driven lending decisions

Student Software Engineer

Aug. 2023 – Feb. 2025

Monash Connected Autonomous Vehicle

Melbourne, VIC

- Implemented ROS 2 launch systems in Python and C++ to synchronise 8 LiDAR sensors on a Nissan e-NV200 van, achieving deterministic and real-time sensor fusion for robust autonomy of the van
- Developed an autonomous valet parking system using the Autoware (ROS 2) stack for sub-50ms control latency

COMMUNITY AND LEADERSHIP

Mental Health Immersive Technologies Developer

GitHub | Feb. 2026 – Apr. 2026

EUGLOH

Paris, France

- Integrated TouchDesigner modules, Python scripts and an Arduino microcontroller to develop a Spatial Augmented Reality (SAR) environment bridging art therapy and mental health for a program at University of Paris-Saclay

University Open Day Student Ambassador

Aug. 2024 – Aug. 2025

Monash University

Melbourne, VIC

- Represented the Faculty of Information Technology as a student ambassador at outreach events such as Open Day, guiding 300+ prospective students and their families through degree offerings, job prospects and university life

PROJECTS

All about the Sky | *Next.js, React, TypeScript, Tailwind CSS, HTML, Open-Meteo API*

Demo | **GitHub**

- Built a weather-driven animated sky with React, TypeScript, and the Canvas 2D API, mapping live Open-Meteo data to procedural clouds, rain, and day/night cycles, with mouse and touch interaction via Pointer Events
- Reduced upstream weather API calls to at most 4 per hour per 11 km area by building a cached Next.js route that proxies Open-Meteo, validates inputs, rounds coordinates for privacy, and caches for 15 minutes

Avian Annotator | *React, Spring Boot, Node.js, PostgreSQL, AWS S3, Docker*

GitHub

- Oversaw the backend development of an accessible web-based bird image annotation platform allowing ornithology researchers and citizen scientists to collaboratively create high-quality training datasets for AI models
- Designed and deployed RESTful APIs using Spring Boot and PostgreSQL to efficiently query and manage multi-user annotation data, while integrating AWS S3 for secure, scalable image storage

TECHNICAL SKILLS

Languages: Python, C++, Go, Java, JavaScript, TypeScript, SQL, HTML, CSS

Frameworks/Libraries: React, Node.js, Spring Boot, FastAPI, Flask, PyTorch, Bootstrap, ROS 2, Tailwind CSS

Mathematics/Physics: Linear Algebra, Probability, Statistics, Graph Theory, Calculus, Fluid Mechanics